4-2 & 3-2 Directional Control Valves

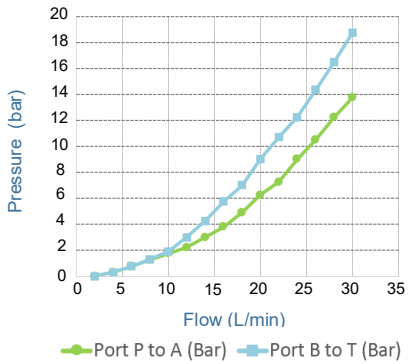
Selection data & performance | Manufacturer technical reference.

A patented range of valves which control the directional flow of corrosive fluids in hydraulic systems. Designed specifically for fluids with low viscosity, the valves offer high efficiency with minimal energy input for actuation.
Available in 4/2 & 3/2 control functions, the rest position in both normally open or normally closed are also offered. Various porting configurations including stand alone (BSPP), TWHC manifold or Cetop 3 are available. Please see manifolds data sheet for options and cavity details.

SPECIFICATION	
Max Inlet Pressure	160 Bar
Operating Pressure Range	0 to 160 Bar
Flow Range	0-30 L/min
Flow Characteristics	Cv Rate 0.62 Kv Rate 0.75
Feed Gallery Diameter	6mm
Porting	1/4" BSP (Parallel)
Construction Materials	A.I.S.I 316/Polymer
Solenoid Operation	24V DC
Media	Water
Electrical Connections	Spade or Hirschmann



Flow Characteristics

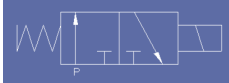
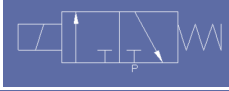


PWM Circuit

For direct acting solenoid actuation above 20 bar, actuation card (part number: 004-010) must be used. This minimises the electrical power consumption whilst at the same time maximising the actuation force of the solenoid. For power requirement and wiring instructions please see actuation card data sheet.

Actuation

Solenoid actuation has a manual over ride as standard. Actuation options available are hand, detent, air pilot, water pilot and oil pilot. We also offer a combination of actuation and return; please contact TWHC for further information.

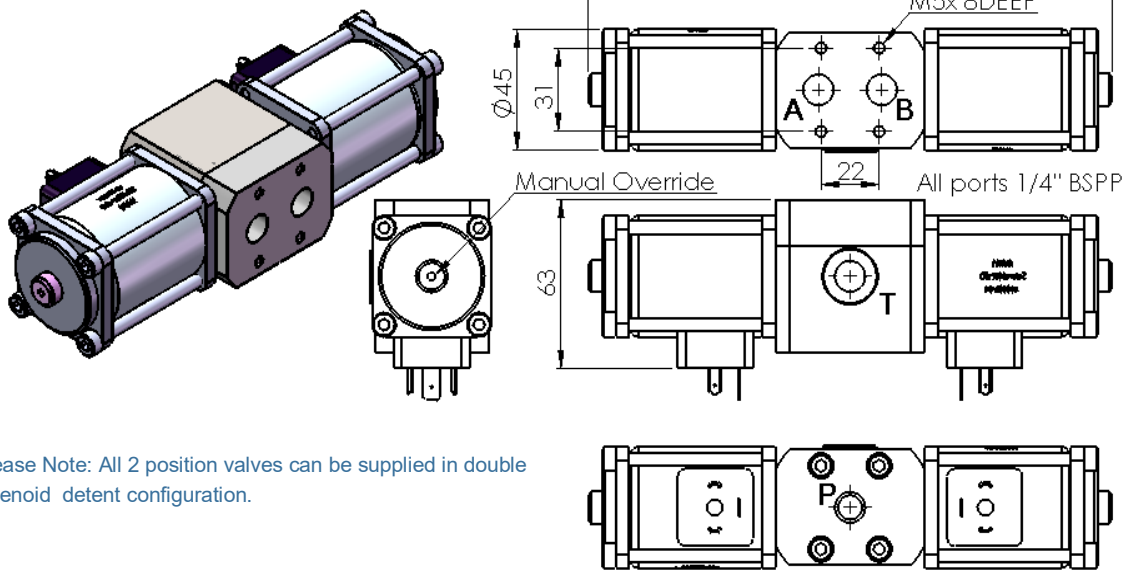
ORDERING CODES	STANDARD BUILD	MANIFOLD MOUNT	CETOP MOUNT
 4/2 Valve	201BBSW	2M01BBSW	2C01BBSW
 3/2 Valve Spring to Open P	208BBSW	2M08BBSW	2C08BBSW
 3/2 Valve Spring to Closed P	203BBSW	2M03BBSW	2C03BBSW

Match the spool function, actuator flow and switching sequence. Include return pressure and define the state required after loss of power. Water service removes hydraulic oil from the fluid circuit. Minimise unnecessary pressure loss; request configuration-specific approval evidence.

4-2 & 3-2 Directional Control Valves

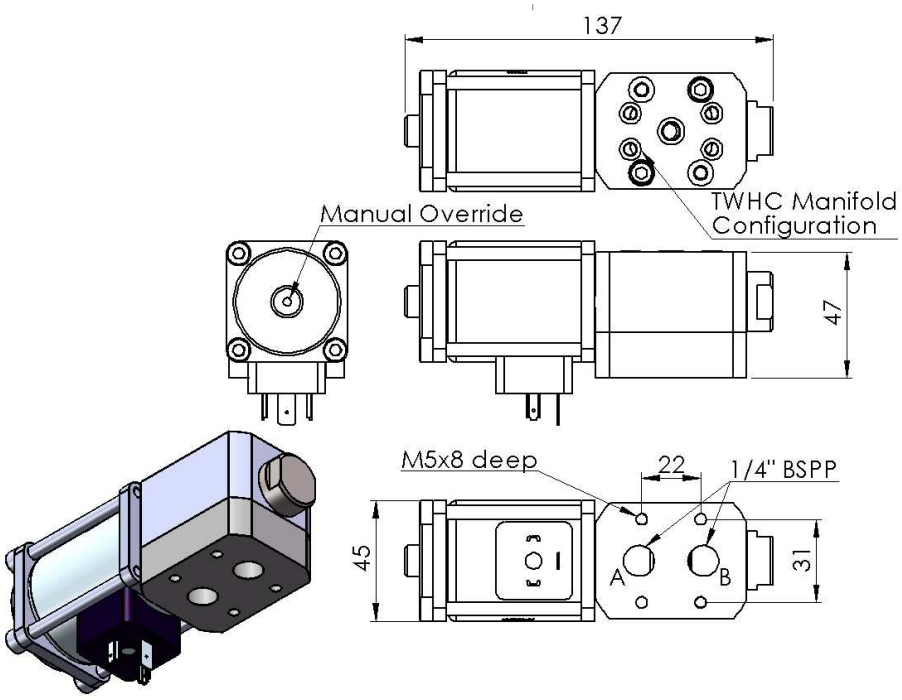
Dimensions, connections & configuration | Manufacturer technical reference.

201BBBW



Please Note: All 2 position valves can be supplied in double solenoid detent configuration.

2M01BBSW



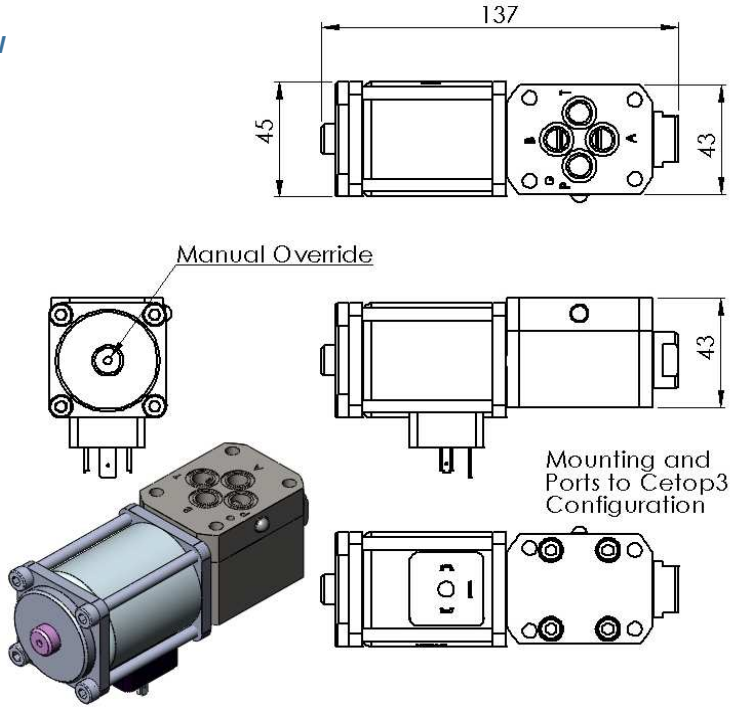
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4-2 & 3-2 Directional Control Valves V.01.25.pdf, p. 2. Manufacturer labels and dimensions retained.

4-2 & 3-2 Directional Control Valves

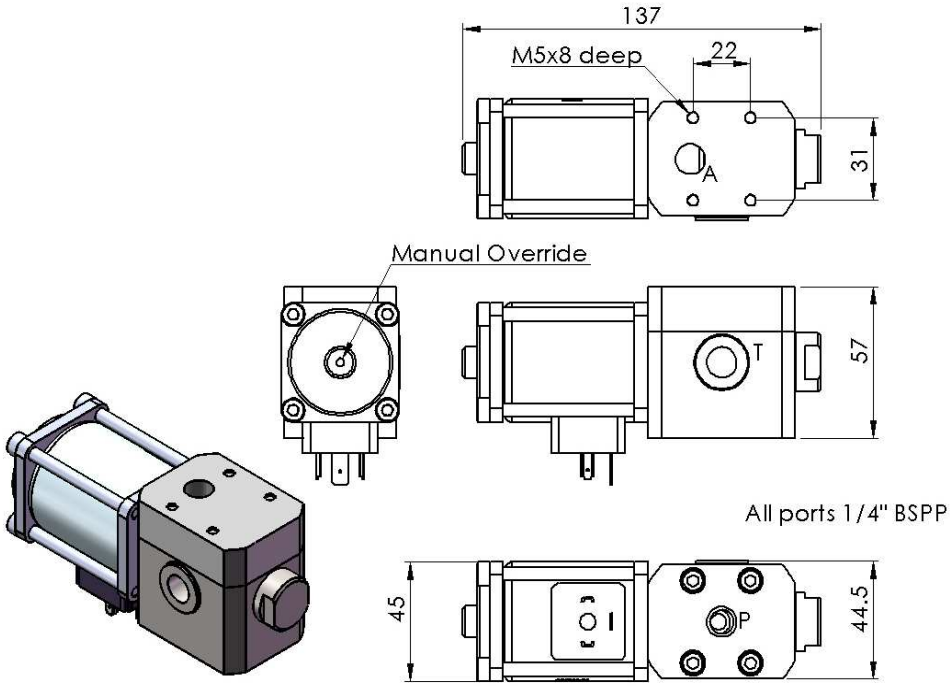
Dimensions, connections & configuration | Manufacturer technical reference.

2C01BBSW, 2C03BBSW & 2C08BBSW



Please Note: The sprung position is port P joined to port B, port A joined to port T.

208BBSW



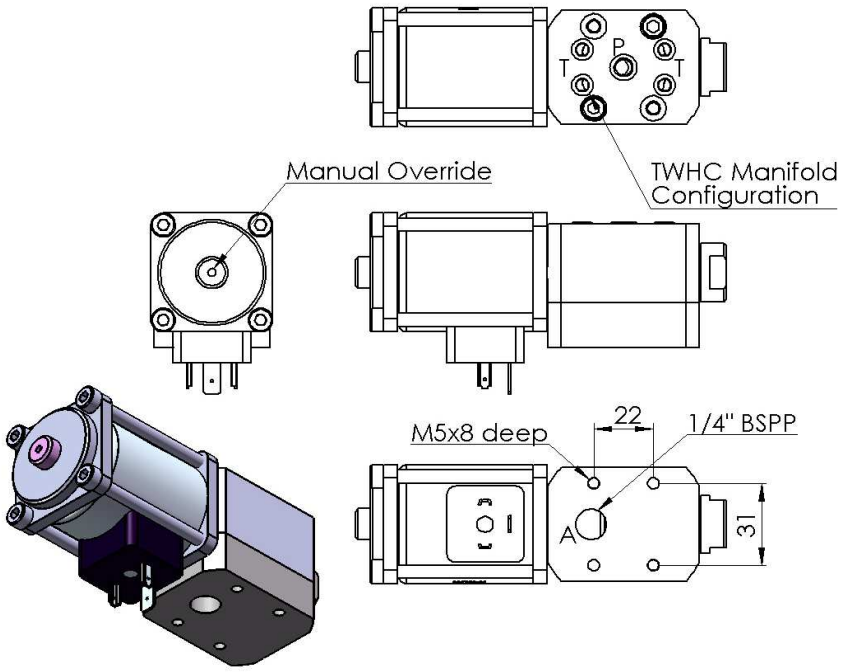
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4-2 & 3-2 Directional Control Valves V.01.25.pdf, p. 3. Manufacturer labels and dimensions retained.

4-2 & 3-2 Directional Control Valves

Dimensions, connections & configuration | Manufacturer technical reference.

2M08BBSW



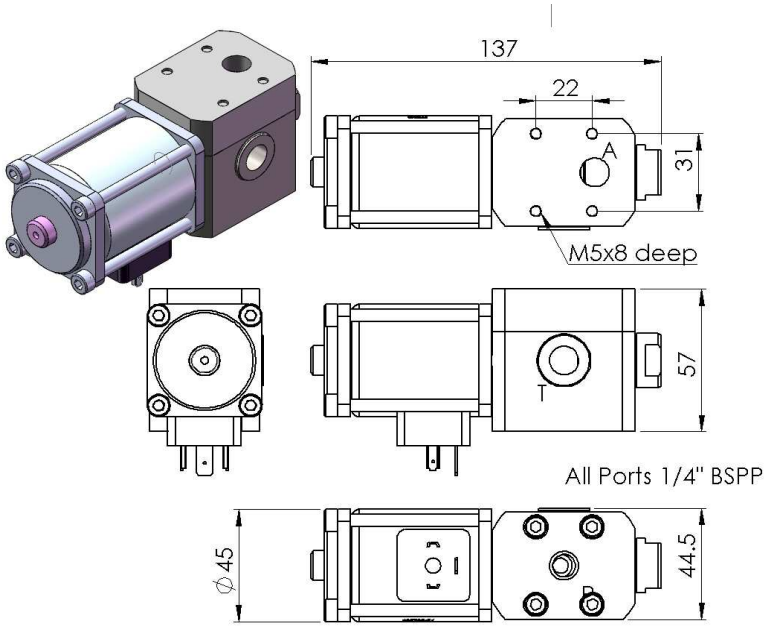
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4-2 & 3-2 Directional Control Valves V.01.25.pdf, p. 4. Manufacturer labels and dimensions retained.

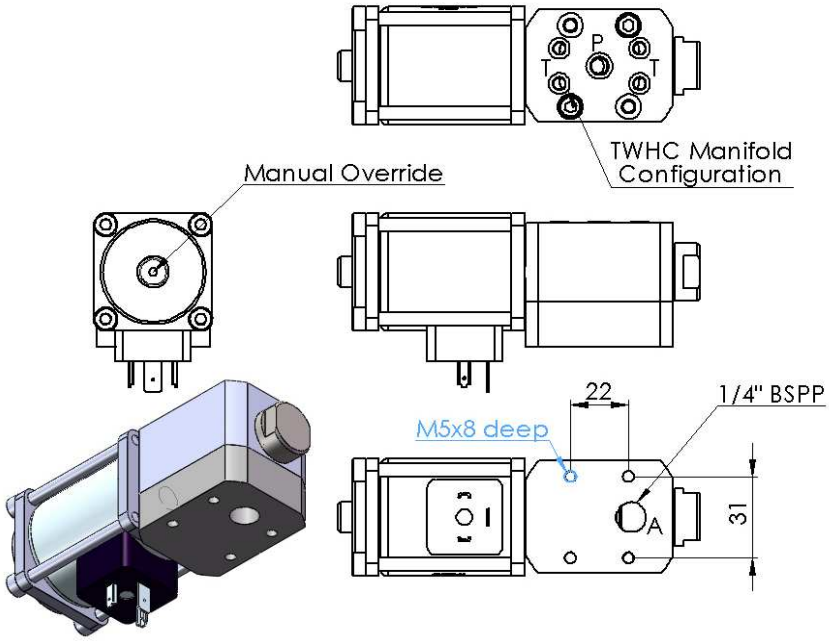
4-2 & 3-2 Directional Control Valves

Dimensions, connections & configuration | Manufacturer technical reference.

203BBSW



2M03BBSW



Please Note: All system components must be manufactured in non-corrosive materials; i.e. Stainless Steel or plastic where suitable. Ferrous particles & oxides are very aggressive and will seriously damage the health of your Water Hydraulics product. High wear; erratic performance and premature failure will result if this is not adhered to.

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FM 87247

Match the spool function, actuator flow and switching sequence. Include return pressure and define the state required after loss of power. Water service removes hydraulic oil from the fluid circuit. Minimise unnecessary pressure loss; request configuration-specific approval evidence.

4-2 & 3-2 Directional Control Valves V.01.25.pdf, p. 5. Manufacturer labels and dimensions retained.